

EXTENSION 2

Stage 6

Vectors (Ext2), V1 Working With Vectors (Ext2)

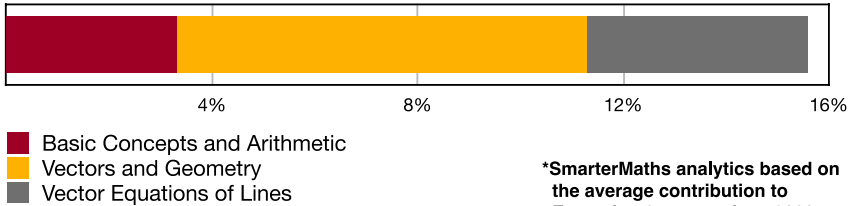
Vectors and Vector Equations of Lines

Teacher: Nardene Dodd

Exam Equivalent Time: 93 minutes (based on allocation of 1.5 minutes per mark)



V1 Working with Vectors



***SmarterMaths analytics based on the average contribution to Extension 2 exams since 2020.**

HISTORICAL CONTRIBUTION

- V1 Working With Vectors has contributed an impressive 15.5% per new syllabus Ext2 exam since it was introduced in 2020.
- We have split this topic into three categories for analysis purposes which are: 1-Basic Concepts and Arithmetic (3.3%), 2-Vectors and Geometry (8.0%) and 3-Vectors and Vector Equations of Lines (4.2%).
- This analysis looks at Vectors and Vector Equations of Lines.

HSC ANALYSIS - What to expect and common pitfalls

- Vectors and Vector Equations of Lines (4.2%) has appeared in the longer answer section of every new syllabus exam so far, as well as further multiple choice questions in 2023, 2021 (twice) and 2020.
- Competency in calculations relating to parallel vectors (2023, 2022), perpendicular vectors (2021) and vector intersection (2020) is required. While most are low difficulty examiner gifts worth 2–3 marks, 2023 Ext2 5 MC was poorly answered and deserves attention.
- Finding the equation of a line between 2 points appeared in 2023 Ext2 11c and 2021 Ext2 3 MC. A core skill that must be revised.
- Converting from vector to Cartesian form (and less often vice versa) is an important skill that has already been required in 2 separate questions (2020 Ext2 3 MC and 2022 Ext2 11e).
- Identifying the position vector of a curve was a challenging multiple choice in 2021 Ext2 7 MC and deserves revision attention.
- A deep understanding of this concept can also be applied to more difficult Vectors and Geometry questions of which 2022 Ext2 14a is case in point.

Questions

1. Vectors, EXT2 V1 2021 HSC 3 MC

Which of the following is a vector equation of the line joining the points **A(4, 2, 5)** and **B(−2, 2, 1)**?

- A.

$$\vec{r} = \begin{pmatrix} 4 \\ 2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$
- B.

$$\vec{r} = \begin{pmatrix} 4 \\ 2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$$
- C.

$$\vec{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 2 \\ 5 \end{pmatrix}$$
- D.

$$\vec{r} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 2 \\ 5 \end{pmatrix}$$

2. Vectors, EXT2 V1 2020 HSC 3 MC

What is the Cartesian equation of the line $\vec{r} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 4 \end{pmatrix}$?

- A.

$$2y + x = 7$$
- B.

$$y - 2x = - 5$$
- C.

$$y + 2x = 5$$
- D.

$$2y - x = - 1$$

3. Vectors, EXT1 V1 SM-Bank 11 MC

Which pair of line segments intersect at exactly one point

A.
$$\begin{cases} \vec{u} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \end{pmatrix}, & 0 \leq \lambda \leq 1 \\ \vec{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -4 \end{pmatrix}, & 0 \leq \lambda \leq 1 \end{cases}$$

B.
$$\begin{cases} \vec{u} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -1 \end{pmatrix}, & 0 \leq \lambda \leq 1 \\ \vec{v} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \end{pmatrix}, & 0 \leq \lambda \leq 1 \end{cases}$$

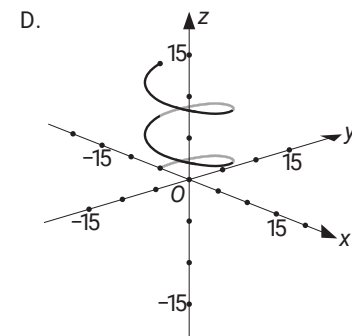
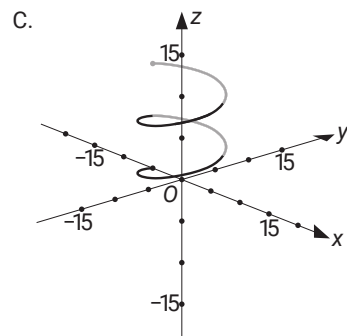
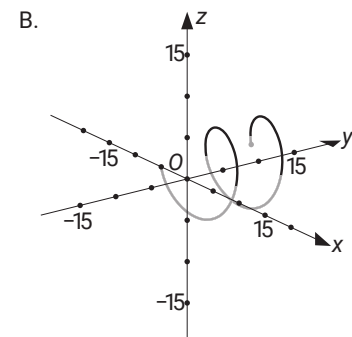
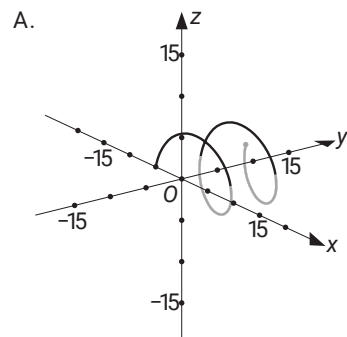
C.
$$\begin{cases} \vec{u} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 6 \end{pmatrix}, & 0 \leq \lambda \leq 1 \\ \vec{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \end{pmatrix}, & 0 \leq \lambda \leq 1 \end{cases}$$

D.
$$\begin{cases} \vec{u} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \end{pmatrix}, & 0 \leq \lambda \leq 1 \\ \vec{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \end{pmatrix}, & 0 \leq \lambda \leq 1 \end{cases}$$

4. Vectors, EXT2 V1 2021 HSC 7 MC

Which diagram best shows the curve described by the position vector

$$\vec{r}(t) = -5\cos(t)\vec{i} + 5\sin(t)\vec{j} + t\vec{k} \text{ for } 0 \leq t \leq 4\pi?$$



5. Vectors, EXT2 V1 2023 HSC 5 MC

Which of the following is a true statement about the lines $\ell_1 = \begin{pmatrix} -1 \\ 2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}$ and

$$\ell_2 = \begin{pmatrix} 3 \\ -10 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -3 \\ -1 \end{pmatrix}?$$

- A. ℓ_1 and ℓ_2 are the same line.
- B. ℓ_1 and ℓ_2 are not parallel and they intersect.
- C. ℓ_1 and ℓ_2 are parallel and they do not intersect.
- D. ℓ_1 and ℓ_2 are not parallel and they do not intersect.

6. Vectors, EXT2 V1 SM-Bank 9

- i. Find the equation of line vector $\tilde{\mathbf{r}}$, given it passes through $(1, 3, -2)$ and $(2, -1, 2)$. (2 marks)
- ii. Determine if $\tilde{\mathbf{r}}$ passes through $(4, -9, 10)$. (1 mark)

7. Vectors, EXT2 V1 2020 HSC 13b

Consider the two lines in three dimensions given by

$$\tilde{\mathbf{r}} = \begin{pmatrix} 3 \\ -1 \\ 7 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \text{ and } \tilde{\mathbf{r}} = \begin{pmatrix} 3 \\ -6 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}.$$

By equating components, find the point of intersection of the two lines. (3 marks)

8. Vectors, EXT2 V1 2021 HSC 12c

Two lines are given by $\mathbf{r}_1 = \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$ and $\mathbf{r}_2 = \begin{pmatrix} 4 \\ -2 \\ q \end{pmatrix} + \mu \begin{pmatrix} p \\ 3 \\ -1 \end{pmatrix}$, where p and q are real numbers. These lines intersect and are perpendicular.
Find the values of p and q . (3 marks)

9. Vectors, EXT2 V1 2022 HSC 11e

Let ℓ_1 be the line with equation $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \lambda \in \mathbb{R}$.
The line ℓ_2 passes through the point $A(-6, 5)$ and is parallel to ℓ_1 .
Find the equation of the line ℓ_2 in the form $y = mx + c$. (2 marks)

10. Vectors, EXT2 V1 2023 HSC 11c

Find a vector equation of the line through the points $A(-3, 1, 5)$ and $B(0, 2, 3)$. (2 marks)

11. Vectors, EXT2 V1 SM-Bank 1

Consider the vectors $\tilde{\mathbf{u}} = a\tilde{\mathbf{i}} - b\tilde{\mathbf{j}} + c\tilde{\mathbf{k}}$ and $\tilde{\mathbf{v}} = \tilde{\mathbf{i}} - 8\tilde{\mathbf{j}} + 4\tilde{\mathbf{k}}$.
Find all possible values of a, b and c if $\tilde{\mathbf{u}}$ is parallel to $\tilde{\mathbf{v}}$ and has a magnitude of 3. (3 marks)

12. Vectors, EXT2 V1 SM-Bank 18

- A sphere is represented by the equation
- $$x^2 - 4x + y^2 + 8y + z^2 - 3z + 2 = 0$$
- i. Determine the centre $\tilde{\mathbf{c}}$ and radius of the sphere. (2 marks)
- ii. Find the vector equation of the sphere. (1 mark)

13. Vectors, EXT2 V1 SM-Bank 21

- i. Find the equation of the vector line $\tilde{\mathbf{v}}$ that passes through $A(5, 2, 3)$ and $B(7, 6, 1)$. (1 mark)
- ii. A sphere has centre $\tilde{\mathbf{c}}$ at $(2, 3, 5)$ and a radius of $5\sqrt{2}$ units.
Find the points where the vector line $\tilde{\mathbf{v}}$ meets the sphere. (3 marks)

14. Vectors, EXT2 V1 SM-Bank 24

Show that the points $A(2, 1, -1)$, $B(4, 2, -3)$ and $C(-4, -2, 5)$ are collinear. (2 marks)

15. Vectors, EXT2 V1 SM-Bank 3

Find the value of n given

$$\tilde{\mathbf{v}} = \begin{pmatrix} 5 \\ 2 \\ n \end{pmatrix} + \lambda_1 \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$$

is perpendicular to

$$\tilde{\mathbf{u}} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 2 \\ n \\ 1 \end{pmatrix}. \quad (2 \text{ marks})$$

16. Vectors, EXT2 V1 SM-Bank 4

Determine the equation of the line vector $\tilde{\mathbf{r}}$, given it passes through the point $(7, 1, 0)$ and is parallel to the line joining $P(2, -1, 2)$ and $Q(3, 4, 1)$. (2 marks)

17. Vectors, EXT2 V1 SM-Bank 6

- i. What vector line equation, $\tilde{\mathbf{r}}$, corresponds to the Cartesian equation
- $$\frac{x+2}{5} = \frac{y-5}{4} \quad (1 \text{ mark})$$
- ii. Express $\tilde{\mathbf{v}}$ in Cartesian form where,
- $$\tilde{\mathbf{v}} = \begin{pmatrix} 1 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad (1 \text{ mark})$$

18. Vectors, EXT2 V1 SM-Bank 15

Consider the two vector line equations

$$\vec{v}_1 = \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}, \quad \vec{v}_2 = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 4 \\ 2 \\ -6 \end{pmatrix}$$

i. Show that $\vec{v}_1$ and $\vec{v}_2$ intersect and determine the point of intersection. (2 marks)

ii. What is the acute angle between the vector lines, to the nearest minute. (2 marks)

19. Vectors, EXT2 V1 SM-Bank 2

i. Find values of a , b , c and d such that $\vec{v} = \begin{pmatrix} a \\ b \end{pmatrix} + 2\begin{pmatrix} c \\ d \end{pmatrix}$ is a vector equation of a line that passes through $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ -3 \end{pmatrix}$. (2 marks)

ii. Determine whether $\vec{u} = \begin{pmatrix} 4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \end{pmatrix}$ is perpendicular to $\vec{v}$. (1 mark)

iii. Express $\vec{u}$ in Cartesian form. (1 mark)

20. Vectors, EXT2 V1 SM-Bank 10

i. Determine the point of intersection of $\vec{a}$ and $\vec{b}$ given.

$$\vec{a} = \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}, \text{ and}$$

$$\vec{b} = \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \quad (2 \text{ marks})$$

ii. Determine if the point $(2, -2, 5)$ lies on $\vec{b}$. (1 mark)

21. Vectors, EXT2 V1 EQ-Bank 8

Classify the triangle formed by joining the points $A(3, 1, 0)$, $B(-2, 4, 3)$ and $C(3, 3, -2)$. (4 marks)

22. Vectors, EXT2 V1 SM-Bank 7

Find the value of x and y , given

$$\vec{r} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} x \\ y \\ -3 \end{pmatrix}$$

and $\vec{r}$ is perpendicular to both $\vec{v}$ and $\vec{w}$, where

$$\vec{v} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \mu_1 \begin{pmatrix} 3 \\ -3 \\ -1 \end{pmatrix} \text{ and } \vec{w} = \begin{pmatrix} -3 \\ 1 \\ 1 \end{pmatrix} + \mu_2 \begin{pmatrix} -4 \\ -1 \\ -2 \end{pmatrix}. \quad (2 \text{ marks})$$

23. Vectors, EXT2 V1 SM-Bank 8

Use the vector form of the linear equations

$$3x - 2y = 4 \text{ and } 3y + 2x - 6 = 0$$

to show they are perpendicular. (3 marks)

24. Vectors, EXT2 V1 2022 HSC 14a

- i. The two non-parallel vectors $\vec{u}$ and $\vec{v}$ satisfy $\lambda\vec{u} + \mu\vec{v} = \vec{0}$ for some real numbers λ and μ .

Show that $\lambda = \mu = 0$. (2 marks)

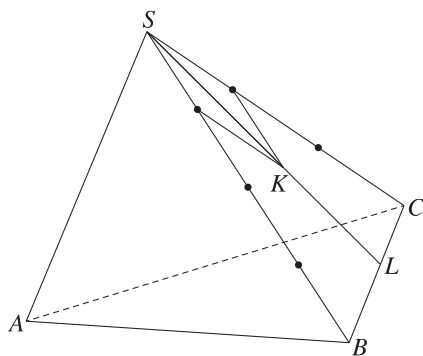
- ii. The two non-parallel vectors $\vec{u}$ and $\vec{v}$ satisfy $\lambda_1\vec{u} + \mu_1\vec{v} = \lambda_2\vec{u} + \mu_2\vec{v}$ for some real numbers $\lambda_1, \lambda_2, \mu_1$ and μ_2 .

Using part (i), or otherwise, show that $\lambda_1 = \lambda_2$ and $\mu_1 = \mu_2$. (1 mark)

The diagram below shows the tetrahedron with vertices A, B, C and S .

The point K is defined by $\vec{SK} = \frac{1}{4}\vec{SB} + \frac{1}{3}\vec{SC}$, as shown in the diagram.

The point L is the point of intersection of the straight lines SK and BC .



- iii. Using part (ii), or otherwise, determine the position of L by showing that $\vec{BL} = \frac{4}{7}\vec{BC}$. (2 marks)

- iv. The point P is defined by $\vec{AP} = -6\vec{AB} - 8\vec{AC}$.

Does P lie on the line AL ? Justify your answer. (2 marks)

Worked Solutions

1. Vectors, EXT2 V1 2021 HSC 3 MC

$$\vec{AB} = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} - \begin{pmatrix} 4 \\ 2 \\ 5 \end{pmatrix} = \begin{pmatrix} -6 \\ 0 \\ -4 \end{pmatrix}$$

$$\vec{r} = \begin{pmatrix} 4 \\ 2 \\ 5 \end{pmatrix} + \lambda_1 \begin{pmatrix} -6 \\ 0 \\ -4 \end{pmatrix}$$

$$= \begin{pmatrix} 4 \\ 2 \\ 5 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$$

$$\Rightarrow B$$

2. Vectors, EXT2 V1 2020 HSC 3 MC

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

$$x = 1 - 2\lambda \Rightarrow \lambda = \frac{1 - x}{2}$$

$$y = 3 + 4\lambda \Rightarrow \lambda = \frac{y - 3}{4}$$

$$\frac{y - 3}{4} = \frac{1 - x}{2}$$

$$y - 3 = 2 - 2x$$

$$y + 2x = 5$$

$$\Rightarrow C$$

3. Vectors, EXT1 V1 SM-Bank 11 MC

Since $\begin{pmatrix} 2 \\ -4 \end{pmatrix} = -2 \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ 6 \end{pmatrix} = -3 \begin{pmatrix} -1 \\ 2 \end{pmatrix}$
 $\Rightarrow A$ and C are parallel lines.

Consider D :

$$3\lambda_1 = \lambda_2 \quad \dots (1)$$

$$2-2\lambda_1 = 1 + \lambda_2 \dots (2)$$

Substitute (1) into (2)

$$2-2\lambda_1 = 1 + 3\lambda_1$$

$$\lambda_1 = \frac{1}{5}$$

$$\lambda_2 = \frac{3}{5}$$

Similarly, λ_1, λ_2 in B can be calculated
 and found to be outside $0 \leq \lambda \leq 1$.
 $\Rightarrow D$

4. Vectors, EXT2 V1 2021 HSC 7 MC

By elimination

Check graph coordinates for specific values of t :

When $t = 4\pi$, $\vec{r} = -5\vec{i} + 0\vec{j} + 4\pi\vec{k}$

$\rightarrow$ Eliminate A and B

When $t = \frac{\pi}{4}$, $\vec{r} = \frac{-5}{\sqrt{2}}\vec{i} + \frac{5}{\sqrt{2}}\vec{j} + \frac{\pi}{4}\vec{k}$

$\rightarrow$ Eliminate C

$\Rightarrow D$

5. Vectors, EXT2 V1 2023 HSC 5 MC

Since $\begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix} = -1 \begin{pmatrix} 1 \\ -3 \\ -1 \end{pmatrix}$, ℓ_1 is parallel to ℓ_2

Test if point $(3, -10, 1)$ lies on ℓ_1 :

i.e. $\exists \lambda$ such that

$$\begin{pmatrix} 3 \\ -10 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}$$

$\lambda = -4$ satisfies equation

$\therefore \ell_1$ and ℓ_2 are the same line.

$\Rightarrow A$

◆ Mean mark 49%.

6. Vectors, EXT2 V1 SM-Bank 9

i. Method 1

Let $A(1, 3, -2)$ and $B(2, -1, 2)$

$$\vec{AB} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ -4 \\ 4 \end{pmatrix}$$

$$\vec{r} = \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -4 \\ 4 \end{pmatrix}$$

Method 2

$$\vec{BA} = \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \\ -4 \end{pmatrix}$$

$$\vec{r} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 4 \\ -4 \end{pmatrix}$$

ii. If $(4, -9, 10)$ lies on the vector line,

$\exists \lambda$ that satisfies:

$$1 + \lambda = 4 \quad \dots (1)$$

$$3 - 4\lambda = -9 \quad \dots (2)$$

$$-2 + 4\lambda = 10 \quad \dots (3)$$

$\lambda = 3$ satisfies all equations

$\therefore (4, -9, 10)$ lies on the line.

7. Vectors, EXT2 V1 2020 HSC 13b

$$\begin{aligned}\vec{r_1} &= \begin{pmatrix} 3 \\ -1 \\ 7 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 + \lambda_1 \\ -1 + 2\lambda_1 \\ 7 + \lambda_1 \end{pmatrix} \\ \vec{r_2} &= \begin{pmatrix} 3 \\ -6 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 - 2\lambda_2 \\ -6 + \lambda_2 \\ 2 + 3\lambda_2 \end{pmatrix}\end{aligned}$$

Intersection occurs when:

$$3 + \lambda_1 = 3 - 2\lambda_2 \dots (1)$$

$$-1 + 2\lambda_1 = -6 + \lambda_2 \dots (2)$$

$$7 + \lambda_1 = 2 + 3\lambda_2 \dots (3)$$

Subtract (3)−(1):

$$4 = -1 + 5\lambda_2$$

$$\lambda_2 = 1$$

Substitute $\lambda_2 = 1$ into (1):

$$3 + \lambda_1 = 1$$

$$\lambda_1 = -2$$

Test that $\lambda_1 = -2$, $\lambda_2 = 1$ satisfies (2):

$$-1 + 2 \times -2 = -6 + 1$$

$$-5 = -5$$

$\exists \lambda_1, \lambda_2$, that satisfy all equations

$\Rightarrow$ 3 lines intersect at a point

$\therefore$ Point of intersection

$$= \begin{pmatrix} 3 \\ -6 \\ 2 \end{pmatrix} + 1 \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -5 \\ 5 \end{pmatrix}$$

8. Vectors, EXT2 V1 2021 HSC 12c

Since vectors are perpendicular

$$\begin{aligned}\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} p \\ 3 \\ -1 \end{pmatrix} &= 0 \\ p - 2 &= 0 \\ p &= 2\end{aligned}$$

Since lines intersect, equate y -coordinates:

$$1 + \lambda 0 = -2 + 3\mu$$

$$\mu = 1$$

Find λ by equating x -coordinates:

$$-2 + \lambda = 4 + 1 \times 2$$

$$\lambda = 8$$

Equating z -coordinates:

$$3 + 8 \times 2 = q - 1 \times 1$$

$$q = 20$$

9. Vectors, EXT2 V1 2022 HSC 11e

$$m_{\ell_1} = \frac{2}{3}$$

Equation of ℓ_2 has $m = \frac{2}{3}$ and passes through $(-6, 5)$:

$$y - 5 = \frac{2}{3}(x + 6)$$

$$y = \frac{2}{3}x + 9$$

10. Vectors, EXT2 V1 2023 HSC 11c

$$\vec{AB} = \begin{pmatrix} 0 - (-3) \\ 2 - 1 \\ 3 - 5 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix}$$

$$\therefore \vec{v} = \begin{pmatrix} -3 \\ 1 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix}, \text{ for some } \lambda \in \mathbb{R}$$

11. Vectors, EXT2 V1 SM-Bank 1

$$|\vec{v}| = \sqrt{1 + 64 + 16} = 9$$

$$\hat{v} = \frac{\vec{v}}{|\vec{v}|} = \frac{1}{9}\vec{i} - \frac{8}{9}\vec{j} + \frac{4}{9}\vec{k} \text{ (magnitude of 1)}$$

Since $\vec{u}$ has a magnitude of 3:

$$\begin{aligned}\vec{u} &= \pm 3 \left(\frac{1}{9}\vec{i} - \frac{8}{9}\vec{j} + \frac{4}{9}\vec{k} \right) \\ &= \pm \left(\frac{1}{3}\vec{i} - \frac{8}{3}\vec{j} + \frac{4}{3}\vec{k} \right)\end{aligned}$$

$$\therefore a, b, c = \frac{1}{3}, -\frac{8}{3}, \frac{4}{3} \text{ or } -\frac{1}{3}, \frac{8}{3}, -\frac{4}{3}$$

12. Vectors, EXT2 V1 SM-Bank 18

i. $x^2 - 4x + y^2 + 8y + z^2 - 3z + 2 = 0$

$$(x-2)^2 + (y+4)^2 + \left(z - \frac{3}{2}\right)^2 + 2 - \frac{89}{4} = 0$$

$$(x-2)^2 + (y+4)^2 + \left(z - \frac{3}{2}\right)^2 = \frac{81}{4}$$

$$\therefore \vec{c} = \begin{pmatrix} 2 \\ -4 \\ \frac{3}{2} \end{pmatrix}, \text{ radius} = \frac{9}{2}$$

ii. Vector equation:

$$\left| \vec{r} - \begin{pmatrix} 2 \\ -4 \\ \frac{3}{2} \end{pmatrix} \right| = \frac{9}{2}$$

13. Vectors, EXT2 V1 SM-Bank 21

i. $\vec{BA} = \begin{pmatrix} 7-5 \\ 6-2 \\ 1-3 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ -2 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$

$$\vec{v} = \begin{pmatrix} 5 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

ii. General point $\vec{v}$:

$$x = 5 + \lambda$$

$$y = 2 + 2\lambda$$

$$z = 3 - \lambda$$

Equation of sphere, $\vec{c} = (2, 3, 5)$, radius $5\sqrt{2}$:

$$(x-2)^2 + (y-3)^2 + (z-5)^2 = (5\sqrt{2})^2$$

$$(\lambda + 3)^2 + (2\lambda - 1)^2 + (-\lambda - 2)^2 = 50$$

$$\lambda^2 + 6\lambda + 9 + 4\lambda^2 - 4\lambda + 1 + \lambda^2 + 4\lambda + 4 = 50$$

$$6\lambda^2 + 6\lambda + 14 = 50$$

$$6\lambda^2 + 6\lambda - 36 = 0$$

$$6(\lambda + 3)(\lambda - 2) = 0$$

$$\lambda = -3 \text{ or } 2$$

When $\lambda = -3$,

$$\text{Intersection} = \begin{pmatrix} 5 \\ 2 \\ 3 \end{pmatrix} - 3 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix}$$

When $\lambda = 2$,

$$\text{Intersection} = \begin{pmatrix} 5 \\ 2 \\ 3 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 7 \\ 6 \\ 1 \end{pmatrix}$$

14. Vectors, EXT2 V1 SM-Bank 24

$$\begin{aligned}\overrightarrow{AB} &= \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \\ \overrightarrow{AC} &= \begin{pmatrix} -4 \\ -2 \\ 5 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -6 \\ -3 \\ 6 \end{pmatrix} = -3 \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}\end{aligned}$$

Since $\overrightarrow{AB} \parallel \overrightarrow{AC}$ and A lies on both lines,
 A, B, C are collinear.

15. Vectors, EXT2 V1 SM-Bank 3

Given $\vec{v} \perp \vec{u}$:

$$\begin{aligned}\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ n \\ 1 \end{pmatrix} &= 0 \\ 4 + n + 3 &= 0 \\ n &= -7\end{aligned}$$

16. Vectors, EXT2 V1 SM-Bank 4

$$\begin{aligned}\overrightarrow{PQ} &= \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ -1 \end{pmatrix} \\ \therefore \vec{r} &= \begin{pmatrix} 7 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \\ -1 \end{pmatrix}\end{aligned}$$

17. Vectors, EXT2 V1 SM-Bank 6

$$\text{i. } \vec{r} = \begin{pmatrix} -2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 4 \end{pmatrix}$$

$$\text{ii. } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$x = 1 + 3\lambda \Rightarrow \lambda = \frac{x-1}{3}$$

$$y = -4 + \lambda \Rightarrow \lambda = y + 4$$

$$y + 4 = \frac{x-1}{3}$$

$$\therefore y = \frac{x-13}{3}$$

COMMENT: Ensure you know the format for conversion from Cartesian to vector form (part i)!

18. Vectors, EXT2 V1 SM-Bank 15

i. Solve simultaneously:

$$1 + 3\lambda_1 = 3 + 4\lambda_2 \quad \dots (1)$$

$$4 + 0\lambda_1 = 2 + 2\lambda_2 \quad \dots (2)$$

$$-2 - \lambda_1 = 2 - 6\lambda_2 \quad \dots (3)$$

$$\Rightarrow \lambda_2 = 1 \quad (\text{from (2)})$$

$$\Rightarrow \lambda_1 = 2 \quad (\text{from (1) and (3)})$$

$\therefore$ vector lines intersect

$$\text{P.O.I.} = \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix} + 2 \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 7 \\ 4 \\ -4 \end{pmatrix}$$

$$\text{ii. } \begin{aligned}\vec{v}_1 &= \vec{a}_1 + \lambda_1 \cdot \vec{b}_1 \\ \vec{v}_2 &= \vec{a}_2 + \lambda_2 \cdot \vec{b}_2\end{aligned}$$

$$\begin{aligned}\cos\theta &= \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1| |\vec{b}_2|} \\ &= \frac{12 + 0 + 6}{\sqrt{10}\sqrt{56}} \\ &= 0.7606\dots\end{aligned}$$

$$\theta = 40.479\dots$$

$$= 40^\circ 29' \quad (\text{nearest minute})$$

19. Vectors, EXT2 V1 SM-Bank 2

i. Method 1

$$\overrightarrow{OA} = \underset{\sim}{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad \overrightarrow{OB} = \underset{\sim}{b} = \begin{pmatrix} -3 \\ -3 \end{pmatrix}$$

$$\begin{aligned} \overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} \\ &= \begin{pmatrix} -3 \\ -3 \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} -6 \\ -4 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \underset{\sim}{v} &= \underset{\sim}{a} + \lambda \underset{\sim}{b} \\ &= \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -6 \\ -4 \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} -3 \\ -2 \end{pmatrix} \end{aligned}$$

$$\therefore a = 3, b = 1, c = -3, d = -2$$

Method 2

$$\begin{aligned} \overrightarrow{BA} &= \overrightarrow{OA} - \overrightarrow{OB} \\ &= \begin{pmatrix} 3 \\ 1 \end{pmatrix} - \begin{pmatrix} -3 \\ -3 \end{pmatrix} \\ &= \begin{pmatrix} 6 \\ 4 \end{pmatrix} \end{aligned}$$

$$\underset{\sim}{v} = \begin{pmatrix} -3 \\ -3 \end{pmatrix} + 2 \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$\therefore a = -3, b = -3, c = 3, d = 2$$

$$\text{ii. } \underset{\sim}{u} = \begin{pmatrix} 4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

$$\begin{aligned} \underset{\sim}{v} &= \begin{pmatrix} 3 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} -3 \\ -2 \end{pmatrix} \\ \begin{pmatrix} -2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ -2 \end{pmatrix} &= 6 - 6 = 0 \\ \therefore \underset{\sim}{u} &\perp \underset{\sim}{v} \end{aligned}$$

$$\text{iii. } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

$$x = 4 - 2\lambda \quad \dots (1)$$

$$y = 6 + 3\lambda \quad \dots (2)$$

Substitute $\lambda = \frac{4-x}{2}$ from (1) into (2):

$$y = 6 + 3 \left(\frac{4-x}{2} \right)$$

$$y = 6 + 6 - \frac{3x}{2}$$

$$y = -\frac{3}{2}x + 12$$

20. Vectors, EXT2 V1 SM-Bank 10

i. At point of intersection:

$$3 + \lambda = -2 + \mu \quad \dots (1)$$

$$5 + 3\lambda = 2 - \mu \quad \dots (2)$$

$$1 - 2\lambda = -1 + 2\mu \quad \dots (3)$$

$$(1) + (2)$$

$$8 + 4\lambda = 0$$

$$\lambda = -2, \quad \mu = 3$$

Intersection (using $\lambda = -2$):

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 - 2 \times 1 \\ 5 - 2 \times 3 \\ 1 - 2 \times -2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 5 \end{pmatrix}$$

ii. If $(2, -2, -10)$ lies on $\underset{\sim}{b}$, $\exists \mu$ that satisfies:

$$-2 + \mu = 2 \quad \dots (1) \Rightarrow \mu = 4$$

$$2 - \mu = -2 \quad \dots (2) \Rightarrow \mu = 4$$

$$-1 + 2\mu = 5 \quad \dots (3) \Rightarrow \mu = 3$$

$\Rightarrow$ No solution

$\therefore (2, -2, 5)$ does not lie on $\underset{\sim}{b}$.

21. Vectors, EXT2 V1 EQ-Bank 8

Calculating the side lengths:

$$|AB| = \sqrt{(3+2)^2 + (1-4)^2 + (0-3)^2} = \sqrt{43}$$

$$|BC| = \sqrt{(-2-3)^2 + (4-3)^2 + (3+2)^2} = \sqrt{51}$$

$$|AC| = \sqrt{(3-3)^2 + (1-3)^2 + (0+2)^2} = \sqrt{8}$$

 $\Rightarrow$ Triangle is scalene.

From above,

$$|BC|^2 = |AB|^2 + |AC|^2 \quad (\angle BAC = 90^\circ)$$

Consider scalar product of direction vectors:

$$\overrightarrow{AB} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2-3 \\ 4-1 \\ 3-0 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 3 \\ 3 \end{pmatrix}$$

$$\overrightarrow{AC} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 3-3 \\ 3-1 \\ -2-0 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix}$$

$$\overrightarrow{AB} \cdot \overrightarrow{AC} = \begin{pmatrix} -5 \\ 3 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix} = 0 + 6 - 6 = 0$$

$$\therefore \overrightarrow{AB} \perp \overrightarrow{AC}$$

 $\therefore \triangle ABC$ is a right-angled (scalene) triangle with a right-angle at A.

22. Vectors, EXT2 V1 SM-Bank 7

Since $\vec{r}$ is perpendicular to $\vec{v}$ and $\vec{w}$:

$$\begin{pmatrix} x \\ y \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -3 \\ -1 \end{pmatrix} = 0$$

$$3x - 3y + 3 = 0$$

$$x - y = -1 \dots (1)$$

$$\begin{pmatrix} x \\ y \\ -3 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ -1 \\ -2 \end{pmatrix} = 0$$

$$-4x - y + 6 = 0$$

$$4x + y = 6 \dots (2)$$

$$(1) + (2)$$

$$5x = 5$$

$$\therefore x = 1$$

$$\therefore y = 2$$

23. Vectors, EXT2 V1 SM-Bank 8

$$3x - 2y = 4$$

$$3x = 2y + 4$$

$$\frac{3}{2}x = y + 2$$

$$\frac{x}{\frac{2}{3}} = y + 2$$

$$\vec{v}_1 = \begin{pmatrix} 0 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} \frac{2}{3} \\ 1 \end{pmatrix}$$

$$3y + 2x - 6 = 0$$

$$2x = -3y + 6$$

$$-\frac{2}{3}x = y - 2$$

$$\frac{x}{-\frac{3}{2}} = y - 2$$

$$\vec{v}_2 = \begin{pmatrix} 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -\frac{3}{2} \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} \frac{2}{3} \\ 1 \end{pmatrix} \begin{pmatrix} -\frac{3}{2} \\ 1 \end{pmatrix} = -1 + 1 = 0$$

$$\therefore \vec{v}_1 \perp \vec{v}_2$$

24. Vectors, EXT2 V1 2022 HSC 14a

i. $\lambda \vec{u} + \mu \vec{v} = \vec{0}$

$$\lambda \vec{u} = -\mu \vec{v}$$

$$\lambda = 0 \text{ or } \vec{u} = -\left(\frac{\mu}{\lambda}\right)\vec{v} = k\vec{v} \quad (k \in \mathbb{R})$$

Since $\vec{u}$ and $\vec{v}$ are not parallel

$$\Rightarrow \lambda = \mu = 0$$

ii. $\lambda_1 \vec{u} + \mu_1 \vec{v} = \lambda_2 \vec{u} + \mu_2 \vec{v}$

$$\lambda_1 \vec{u} - \lambda_2 \vec{u} + \mu_1 \vec{v} - \mu_2 \vec{v} = \vec{0}$$

$$(\lambda_1 - \lambda_2)\vec{u} + (\mu_1 - \mu_2)\vec{v} = \vec{0}$$

Using part (i):

$$(\lambda_1 - \lambda_2) = 0 \text{ and } (\mu_1 - \mu_2) = 0$$

$$\therefore \lambda_1 = \lambda_2 \text{ and } \mu_1 = \mu_2 \dots \text{as required}$$

iii. $\vec{BL} = \lambda \vec{BC}$

$$= \lambda (\vec{BS} + \vec{SC}) \dots (1)$$

$$\vec{BL} = \vec{BS} + \mu \vec{SK}$$

$$= -\vec{SB} + \mu \left(\frac{1}{4}\vec{SB} + \frac{1}{3}\vec{SC} \right)$$

$$= -\vec{SB} + \frac{\mu}{4}\vec{SB} + \frac{\mu}{3}\vec{SC}$$

$$= \frac{\mu}{3}\vec{SC} + \left(\frac{\mu}{4} - 1 \right) \vec{SB} \dots (2)$$

Using (1) = (2):

$$\lambda \vec{BS} + \lambda \vec{SC} = \frac{\mu}{3}\vec{SC} + \left(\frac{\mu}{4} - 1 \right) \vec{SB}$$

$$\frac{\mu}{3} = \lambda, \quad 1 - \frac{\mu}{4} = \lambda$$

$$\frac{\mu}{3} = 1 - \frac{\mu}{4}$$

$$\frac{4\mu + 3\mu}{12} = 1$$

$$\frac{7\mu}{12} = 1$$

$$\mu = \frac{12}{7}$$

Mean mark (ii) 56%.

◆◆◆ Mean mark (iii) 25%.

$$\lambda = \frac{\frac{12}{7}}{3} = \frac{4}{7}$$

$$\therefore \overrightarrow{BL} = \frac{4}{7}\overrightarrow{BC} \dots \text{as required}$$

$$\text{iv. } \overrightarrow{AP} = -6\overrightarrow{AB} - 8\overrightarrow{AC}$$

If P lies on AL , $\exists k$ such that $\overrightarrow{AP} = k\overrightarrow{AL}$

$$-6\overrightarrow{AB} - 8\overrightarrow{AC} = k\overrightarrow{AL}$$

$$-6\overrightarrow{AB} - 8\left(\overrightarrow{AB} + \overrightarrow{BC}\right) = k\left(\overrightarrow{AB} + \overrightarrow{BL}\right)$$

$$-14\overrightarrow{AB} - 8\overrightarrow{BC} = k\left(\overrightarrow{AB} + \frac{4}{7}\overrightarrow{BC}\right) \quad (\text{see part (iii)})$$

$$-14\overrightarrow{AB} - 8\overrightarrow{BC} = k\overrightarrow{AB} + \frac{4k}{7}\overrightarrow{BC}$$

$$k = -14, \quad \frac{4k}{7} = -8 \Rightarrow k = -14$$

$\therefore P$ lies on AL .

◆◆◆ Mean mark (iv) 23%.